多周期事件输入

Cycle T_1 | Cycle T_2 | Cycle T_3 | … | Cycle T_n

① 跨周期信号累积 (SignalAccumulator)

按 (Agent, SignalType, Policy, Boundary, ResourceDigest) 五元组分桶 · 同行为路径信号合并累积

AccumBucket

entries: [(edge_id, causal_weight, detection_confidence, timestamp_ns) ...]

② Noisy-OR 多源证据融合

$S(t) = 1 - \prod (1 - w_i \times s_i \times e^{-\lambda(t-t_i)})$

③ 指数时间衰减

$\lambda = \ln 2 / 300s$ (5min 半衰期)

④ Platt Scaling 概率校准 + ECE 监控

$P(attack) = 1/(1+e^{-(7.0\times S-3.0)})$ · ECE > 0.05 触发在线 SGD 重校准

单周期弱信号不再被丢弃

校准概率 → ClassificationResult.progressive_fused